MTH-418a: Inference-I

2023-2024: II Semester
Mid Semester Examination

Time Allowed: 120 Minutes **Maximum Marks: 50**

NOTE: (i) Start answer of every question on a new page. Moreover, attempt all the parts of a question at one place.
(ii) Answer each question legibly, clearly and concisely. Illegible answers will not be graded.
(ii) For each question, provide details of all the steps involved in arriving at the final answer(s)/conclusion(s) and box your final answer(s)/conclusion(s).

1. Identify TRUE and FALSE statements from the following ten statements. In support of your answers, provide appropriate arguments.

(i) If $|T|$ is a sufficient statistic for $\theta \in \Theta$, then T is also a sufficient statistic for $\theta \in \Theta$.

(ii) If T is a complete statistic, then $e^T - T - 1$ is also a complete statistic.

(iii) If T is a complete statistic and T^2 is a sufficient statistic, then T is a minimal sufficient statistic.

(iv) If T is a minimal sufficient statistic, then $T^3 - 2T^2 + 2T + 1$ is also a minimal sufficient statistic.

(v) Let $x_1 = 1.1, x_2 = 1.5$ and $x_3 = 0.4$ be the observed values of a random sample of size 3 from the pdf

$$f_\theta(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}}, \quad x > 0,$$

where $\theta \in (0, \infty) = \Theta$. Then, the maximum likelihood estimate of θ is 0.5.

(vi) Let $x_1 = -1.5, x_2 = -0.2$ and $x_3 = 0.2$ be the observed values of a random sample of size 3 from the pdf

$$f_\theta(x) = \frac{e^{-|x-\theta|}}{2}, \quad -\infty < x < \infty,$$

where $\theta \in \mathbb{R} = \Theta$. Then, the method of moment estimate of θ is 0.2;

(vii) If δ_1 is the UMVUE of $\psi(\theta), \theta \in \Theta$, and δ_2 is an unbiased estimator of $\psi(\theta)$, then $\text{Var}_\theta(\delta_1) = 2\text{Cov}_\theta(\delta_1, \delta_2)$, $\forall\, \theta \in \Theta$;

(viii) Let X_1, X_2 be a random sample from $\text{U}(0, \theta)$, where $\theta \in (0, \infty) = \Theta$. Let $X_{(1)} \leq X_{(2)}$ be the corresponding order statistics. Using Basu's theorem and the distribution of order statistics, $E_\theta(\frac{X_{(1)}}{X_{(2)}}) = 0.5$, $\forall\, \theta > 0$;

(ix) A minimal sufficient statistic is always complete;

(x) If δ_1 and δ_2 are two UMVUEs of $\psi(\theta)$, then $\text{Var}_\theta(\delta_1 - \delta_2) = 0$, $\forall\, \theta \in \Theta$, and $P_\theta(\delta_1 = \delta_2) = 1$, $\forall\, \theta \in \Theta$. $\boxed{1.5 \times 10{=}15 \text{ Marks}}$

2. Let X_1, X_2 be a random sample from a population having the pdf

$$f_\theta(x) = \frac{1}{\pi} \cdot \frac{1}{1 + (x - \theta)^2}, \quad x \in \mathbb{R},$$

where $\theta \in \mathbb{R} = \Theta$. Show that $\underline{T} = (X_{(1)}, X_{(2)})$ is a minimal sufficient statistic? Is $\underline{T}$ a boundedly complete statistic? $\boxed{6{+}5{=}11 \text{ Marks}}$

3. Let X_1, X_2, X_3 be a random sample of size 3 from the pdf

$$f_\theta(x) = \frac{2x}{\theta^2}, \quad 0 < x < \theta,$$

where $\theta \in (0, \infty) = \Theta$.

(i) Find the MLE of θ;

(ii) Find the MME of θ;

(iii) Using the Rao-Blackwell Theorem, find an estimator based on the complete sufficient statistic $X_{(3)} = \max\{X_1, X_2, X_3\}$ that is better than the MME under the absolute error loss function $L(\theta, a) = |a - \theta|$, $a \in \mathbb{R} = \mathcal{A}, \theta \in \Theta$. $\boxed{4{+}4{+}4{=}12 \text{ Marks}}$

4. Let X be a random variable having the pmf

$$f_\theta(x) = \begin{cases} \theta, & \text{if } x = -1 \\ (1-\theta)^2\theta^x, & \text{if } x = 0, 1, 2, \ldots, \\ 0, & \text{otherwise} \end{cases}$$

where $\theta \in (0, 1) = \Theta$. Show that the UMVUE of $\psi(\theta) = \theta(1 - \theta)^2$ does not exist. $\boxed{12 \text{ Marks}}$